

A primer on Confidence Intervals

UGAIN – Black Belt in Six Sigma – Module 3

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Additional information is available on the notes pages.

Agenda and tentative timing

Subject		Tentative timing (min)
Case example – acceptance sampling		10'
Saved by a confidence interval		20'
How to calculate confidence intervals		30'
Testing hypotheses using confidence intervals		30'
Additional examples		30'
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Case example – acceptance sampling

- A lot of 10000 ceramic tiles is submitted to a single stage acceptance sampling procedure by attributes. An AQL of 2% defectives applies for the lot.
 - The test procedure can be stated, using two hypotheses:
 - H_0 (the status quo): the lot can be accepted for its intended use.
 - H_A (special action): the lot cannot be accepted for its intended use.
 - The hypotheses are being tested at a significance level $\alpha = 5\%$, meaning that a good lot has a probability of 95% to pass the test (i.e., accept H_0).
 - The test is carried out based on a single random sample of size 100. A value of $\alpha = 5\%$, corresponds to a critical value of 4.
 - 4 defectives are found in the sample, i.e., the estimate for the percentage defectives is 4%. Nevertheless the lot is accepted at the chosen significance level.
- Although the procedure on the left is statistically fully sound, it is not so obvious to explain it to the uninitiated:
 - What percentage of defective items can be tolerated in the lot? Answer: 2%.
 - What is our estimate of the percentage defectives in the lot? Answer: 4%.
 - Should we accept the lot for its intended use? Answer: YES!
 - Apart from the non-intuitive nature of the testing procedure, the problem lies in the shortcomings of the point estimation of the percentage defectives:
 - A point estimate is rarely correct. Here: less than 1/3rd of all samples have the same percentage defectives as the lot.
 - A point estimate does not provide any information on the location of the lot percentage defectives. Here: is it higher or lower than 4%?
 - Point estimates are not really consistent. A second attempt will most likely give a different value. Then what?

There is a better way of estimating the lot percentage defectives: confidence intervals.

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The 4% value that we extract from the sample is called a point estimate of the actual percentage of defectives in the lot. You can compare point estimates with the bullet holes on a target (or the holes of the arrows in the bird pecking game). The holes are (in the best case) neatly distributed around the target. This is what estimating (measuring) is in the end: aiming for the target.

Of course, the statistician (data scientist with a modern word) will be expected to come up with an estimate of the percentage of defects in the lot to support the conclusion: the lot is OK or is NOK. Saying that the lot is OK while the estimate of the percentage of defects is twice as high as the maximum allowed, does not exactly contribute to the 'good name and fame' of statistics. Anyone who does not understand the statistical methods will probably not gain any confidence from this.

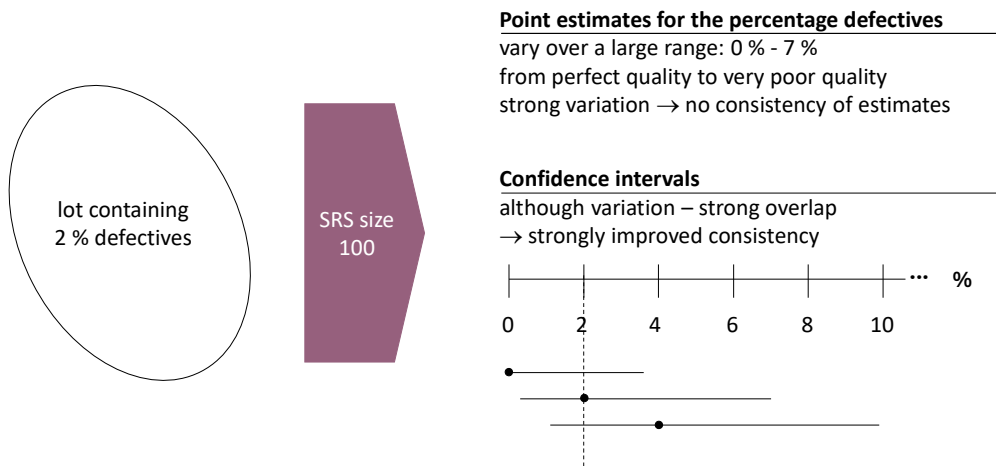
Confidence interval for the lot percentage defectives

- Based on the SRS of size 100 with 4 defectives found, we can state with 95 % confidence that the true percentage defectives of the lot lies between 1.1% and 9.9%.
- Stated in other words: the **95% confidence interval (CI)** for the true percentage defectives is [1.1%, 9.9%].
- The CI provides:
 - A **reliable estimate**: the probability that the true percentage defectives lies in the interval is high: 95%.
 - A **clear** (although not very accurate) **indication where the true percentage defectives is situated**: not lower than 1.1% and not higher than 9.9%.

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How the confidence interval is calculated will be explained further.

Confidence intervals – consistency of the estimates



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Drawn from a lot with 2% defectives, a SRS of size 100, will in practice contain between 0 and 7 defectives, as can be learned from the sampling distribution. When using these point estimates for a judgement call on the quality of the entire lot, then at one extreme we will speak of perfect quality (0% defectives), at the other of absolutely poor quality (7% defectives)

The consistency of the interval estimates is a consequence of the fact that 95% of these intervals must contain the true lot percentage defectives (normally unknown, here 2%). The interval estimates are, as it were, drawn around a fixed value and are therefore overlapping.

You can compare the calculation of confidence intervals with the old folk amusement of throwing horseshoes at a stake driven into the ground. A skilled thrower (and therefore a reliable thrower) will succeed in throwing most of the horseshoes around the stake.

For the sake of completeness, the figures on this slide:
 0 defectives in sample : 95% confidence interval 0,0 % → 3,6 %
 2 defectives in sample : 95% confidence interval 0,3 % → 7,0 %
 4 defectives in sample : 95% confidence interval 1,1 % → 9,9 %

Confidence intervals – confidence and accuracy (1/2)

- A CI (interval estimate) has two important features that a point estimate lacks:
 - The level of **confidence** of the estimate: a design parameter that can be set at any desired value.
 - The **accuracy** of the estimate (the size of the interval): a consequence from the confidence level for a given sample size.
- These are the CIs derived from our sample (size 100 containing 4 defectives) at different levels of confidence:

0,0%	→	[4,0% ; 4,0%]
70,0%	→	[2,1% ; 7,1%]
95,0%	→	[1,1% ; 9,9%]
99,0%	→	[0,7% ; 12,0%]
99,9%	→	[0,4% ; 14,8%]
100,0%	→	[0,0% ; 100,0%]
- From these data we can draw some important conclusions:
 - There is clearly an **inverse relation (trade off) between confidence and accuracy** of the estimate (for a given sample size). The more confident we want to be about our estimate, the less accurate it becomes. The price we pay for 100% confidence is a meaningless estimate. On the other end of the spectrum, we find back our absolute accurate point estimate to which we cannot 'believe'.
 - Note that the CI with confidence level above 70% all contain the AQL value of 2%, while below 70% this is no longer the case. The CI being the range in which we believe the true lot percentage defectives is situated, will hence lead to the acceptance of the hypothesis H_0 that the lot is good for use, when we set our confidence level higher than 70%, but will lead to the rejection of this hypotheses below 70% confidence.

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We find that the intervals become larger as the confidence is increased. If we again make the comparison with horseshoe-throwing: larger (wider) horseshoes are easier to throw around the stake. This means higher confidence (reliability).

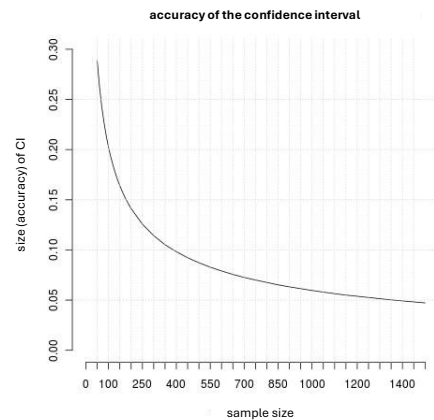
The fact that **for a given sample size** the accuracy decreases when the confidence is increased corresponds to intuition. The information available in the sample must be distributed over both aspects of the estimate, as it were. Statistics allow us to calculate this exactly. In the limit we get perfect confidence, but unfortunately the accuracy is completely lost, making the estimate meaningless. Compare this with the weather forecast that tomorrow there will either be precipitation or that it will remain dry. This is a perfect forecast (100% reliable), but otherwise completely meaningless. You cannot use it to plan your day trip to the seaside.

It is clear by now that CIs offer an alternative for classical hypotheses testing. Interpreting the CI as the range containing the true population value, here the lot percentage defectives π , **it will also be the range of values π_0 for which we will accept H_0 : $\pi = \pi_0$** . The lower the confidence levels we apply, the more we are inclined to reject H_0 hence the higher also the risk of wrongly rejecting H_0 . We can conclude that the level of confidence and the significance of the test provide the same information. We'll come back to this a bit further in this document.

Confidence intervals – confidence and accuracy (2/2)

- Can we have both confidence and accuracy at a high level? YES, however at the expense of an ever-larger sample size.
- It is possible to determine in advance how large the sample needs to be to obtain a desired accuracy in combination with a chosen confidence level.
- From the graph on the right, we deduce that an accuracy of 5% (*) combined with a confidence of 95%, requires a sample with a size of about 1350 elements!

* 5% accuracy means that the size of the interval is 5%, so the actual percentage defectives is determined within a range of 5%. The calculation is based on a sample value of 50%, which is the worst case scenario for the CI size (see further).



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For a given sample size, there is a trade-off between confidence and accuracy. If we want to increase accuracy for a given confidence, we are forced to use a larger sample. Samples can become incredibly large (and therefore lead to a lot of work and expense) when high demands are made on both confidence and accuracy.

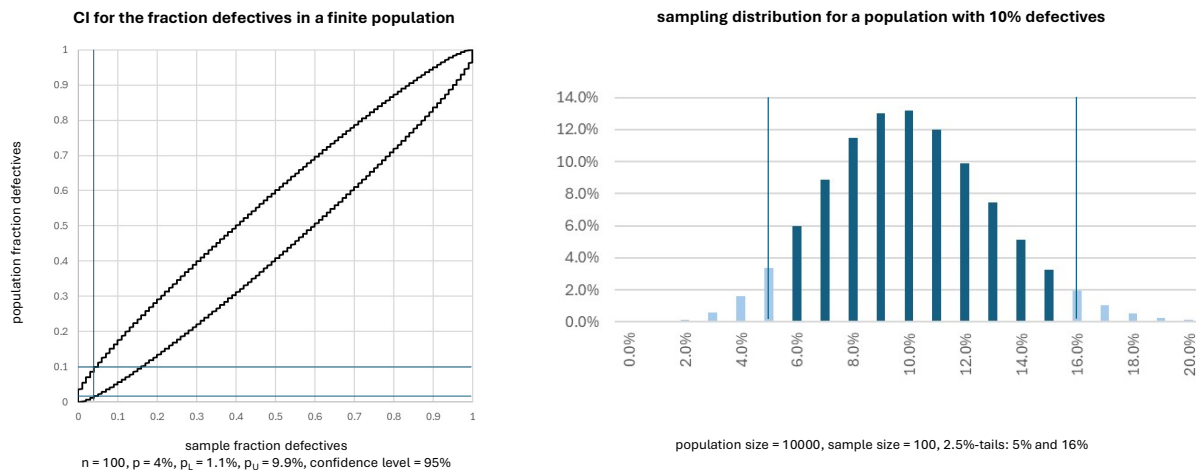
The curve shows the progression of the width of the 95% confidence interval at $p = 50\%$ (the largest in absolute value, so worst case) as a function of sample size. **Roughly speaking, it can be said that doubling the accuracy (this means halving the width of the confidence interval) corresponds to quadrupling the sample size:**

- sample size = 100; width of confidence interval = 20%
- sample size = 400; width of confidence interval = 10%
- sample size = 1350; confidence interval width = 5 %

(The calculations are obviously different when working with relative accuracies. Suppose we want 10 % relative accuracy for an estimate of 10 %. Then this corresponds to a confidence interval width of 2 %.)

Double sided confidence interval

Calculation principle



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Calculating a confidence interval involves a form of function inversion. This means that for a given function $x = f(y)$ (we deliberately write it in this way and not in the usual way $y = f(x)$ to be consistent with the explanation that follows) and a given value x_0 , we have to determine the corresponding value $y_0 = f^{-1}(x_0)$. The figure on the left shows how this works. The (known) percentage of defectives p in the sample is plotted on the horizontal axis (the x-axis), and the (unknown) percentage of defectives π in the lot (population) is plotted on the vertical axis (y-axis). The confidence interval for π that corresponds to the given value for p (here 4%) extends between the two intersections of the curves in the figure with the vertical line through p . To see why this is correct, we need to understand how these two curves (functions) are formed. This is made clear using the figure on the right. This figure shows the sampling distribution corresponding to $\pi = 10\%$. A 'tail probability' of 2.5% has been applied to both ends of this figure (the so-called 2.5% and 97.5% percentiles). The corresponding percentages of defectives (here 5% and 16% respectively) (*) are plotted against π . By repeating this procedure for each value of π , both curves are created. (As π is shown on the y-axis, we have indeed created two functions $x = f(y)$.) It is clearly visible that these curves are not 'smooth', but rather 'saw-toothed'. The reason for this is that both π and p vary in steps of $1/1000$ and $1/100$ respectively and are therefore not continuous quantities. (For π we could have worked with a smaller increment, but this adds little value.) If π_0 is the true percentage of defectives, then 95% of all samples will yield a value for p between the intersections of both curves and the horizontal line through π_0 . **Now it is clear that π_0 lies in the confidence interval for p if and only if p lies between these two intersections. The confidence of the interval (i.e., the probability it contains the true value π_0) is therefore indeed 95%.**

* As we are dealing with discrete quantities, it is not possible to cut off precisely 2.5% tail probabilities. As such the 5% value, is the smallest value for which the tail is at least 2.5%. The same, mutatis mutandis, holds for the 16% value.

From this mechanism it is also immediately clear why a higher confidence yields larger (i.e., less accurate) confidence intervals (for a given sample size). The higher the required level of confidence, the wider the two percentile-functions will be apart.

This mechanism of function inversion can be hard to execute manually (although not always, as we will show on the next slide). A computer however does the job in no time.

Finally note that the largest CI (the least accurate one) is obtained for the value $p = 50\%$, recall that the calculation of the size of the CI as function of the sample size (as shown on an earlier slide) was done for this worst case.

Double sided confidence interval

Easy algebraic function inversion – CI for the population mean

- Given a population quantity X with mean $E[X] = \mu$ and variance $\text{var}[X] = \sigma^2$.
- Central Limit Theorem: for a sufficiently large sample size n , the sample average $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$
 - $\Rightarrow \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1)$
 - $\Rightarrow P\left[t_{0.025, n-1} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_{0.975, n-1}\right] = 95\%$ 95% of values between 2.5% and 97.5% percentiles
 - $\Rightarrow P\left[\bar{X} - t_{0.975, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} - t_{0.025, n-1} \frac{s}{\sqrt{n}}\right] = 95\%$ algebraic reshuffling of formulas
 - $\Rightarrow P\left[\bar{X} - t_{0.975, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{0.975, n-1} \frac{s}{\sqrt{n}}\right] = 95\%$ $t(n-1)$ is a symmetric distribution
- See 'Black Belt in Six Sigma – Confidence Intervals.xlsx' for a demonstration.

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This page shows a case where the function inversion mechanism can be done by simple algebraic calculation. We need the famous Student's t-distribution. This distribution provides us with a relation between the true population mean $E[X] = \mu$ (a fixed value) and the (varying) sample quantities $\bar{X}$ (sample average) and s (sample standard deviation).

The power of the t-distribution lies in the fact that it can be applied for almost any distribution of X , given that the sample size is not too small.

Double sided confidence interval

Easy algebraic function inversion – approximated CI for the population fraction defectives

- Given a lot (population) with (unknown) fraction defectives π
- CLT: for a sufficiently large sample size n , the sample fraction defectives $P \sim N\left(\pi, \sqrt{\frac{\pi(1-\pi)}{n}}\right)$

$$\Rightarrow \frac{P-\pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} \sim Z$$

$$\Rightarrow P\left[Z_{0.025} \leq \frac{P-\pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} \leq Z_{0.975}\right] = 95\%$$

95% of values between 2.5% and 97.5% percentiles

$$\Rightarrow P\left[P - Z_{0.975}\sqrt{\frac{\pi(1-\pi)}{n}} \leq \pi \leq P - Z_{0.025}\sqrt{\frac{\pi(1-\pi)}{n}}\right] = 95\%$$

algebraic reshuffling of formulas

$$\Rightarrow P\left[P - Z_{0.975}\sqrt{\frac{\pi(1-\pi)}{n}} \leq \pi \leq P + Z_{0.975}\sqrt{\frac{\pi(1-\pi)}{n}}\right] = 95\%$$

Z is a symmetric distribution

$$\Rightarrow P\left[P - Z_{0.975}\sqrt{\frac{P(1-P)}{n}} \leq \pi \leq P + Z_{0.975}\sqrt{\frac{P(1-P)}{n}}\right] \cong 95\%$$

approximately

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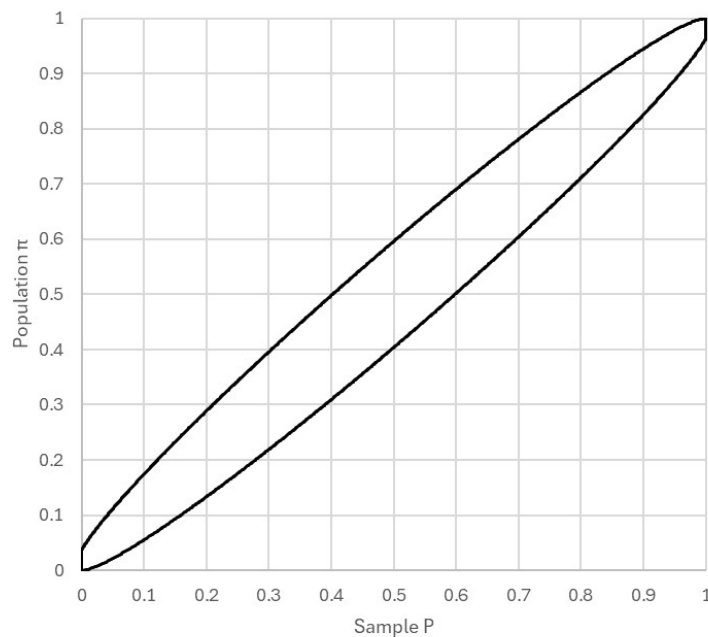
This page shows the famous large sample 95%-CI for the fraction defectives in a lot : $\pi = P \pm 1.96 \sqrt{\frac{P(1-P)}{n}}$. n sufficiently large means that at least 5 defectives need to turn up in the sample. For a defective fraction of 1%, this would mean a sample size of at least 500.

$Z \sim N(0,1)$

$Z_{0.975} = 1.96$

Participants are invited to test this formula with Excel, analogous to the test of the CI for μ mentioned on the previous page.

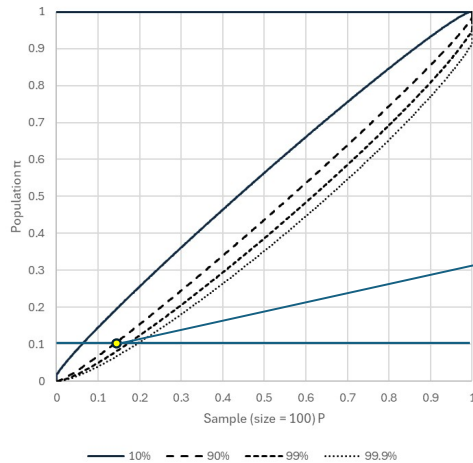
The graph below shows the 2.5% and 97.5% quantile curves for P derived from the normal approximation (sample size = 100).



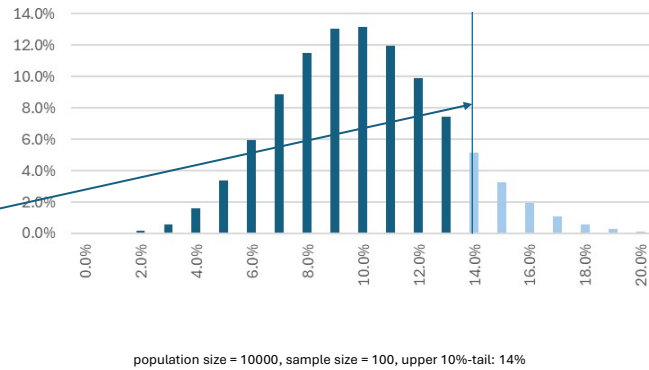
One sided confidence interval

Calculation principle

one sided CI for the fraction defectives in a finite population



sampling distribution for a population with 10% defectives



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Suppose we are only interested in how low the percentage of defectives in a lot could be. In that case, we calculate the one-sided $(1 - \alpha) \times 100\%$ -CI $[\pi_L, 1]$. The calculation principle is the same as for a two-sided CI, but here we only plot the $(1 - \alpha) \times 100\%$ -quantile of the sample distribution. (α usually equal to 5%)

In the same way, the one-sided $(1 - \alpha) \times 100\%$ -CI $[0, \pi_U[$ can be calculated, if we are only interested in how high the percentage of defects in a lot could be.

Also here, due to the discrete nature of the quantities plotted, we will generally not find the exact confidence levels. E.g., the probability of having 14% defectives or more is not exactly 10%, but rather around 12%.

One sided CIs are used for one sided hypotheses tests, as will be demonstrated further.

Confidence interval

Alternative for classical hypothesis testing

$$H_0: \pi \leq \pi_0 \text{ (AQL)}$$

$$H_A: \pi > \pi_0$$

Classical hypothesis test

1. Set significance α
2. Choose sample size n
3. Draw sample and determine test statistic $P = d/n$
4. Determine critical value c for P (or determine p -value)
5. If $P \leq c$ (or p -value $> \alpha$) accept H_0 else reject H_0

Hypothesis test using CI

1. Set confidence level $(1-\alpha) \times 100\%$
2. Choose sample size n
3. Draw sample and determine test statistic $P = d/n$
4. Calculate $(1-\alpha) \times 100\%$ -CI for π : $]\pi_L, 1]$
5. If $\pi_0 \in \text{CI}$ accept H_0 else reject H_0 .

Evidently both test procedures should be equivalent, i.e., they should have the same OC-curve.

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We use the case of the lot fraction defectives to illustrate the general principle. Further in the document we show other examples. When testing a population quantity θ via a CI, we need an estimator for θ as test statistic.

The slide shows the case which is the standard for acceptance sampling, where H_0 corresponds to the desired situation that the fraction defectives is small enough to continue using the lot as intended. The two other cases proceed as follows:

Double sided test for π

$$H_0: \pi = \pi_0 \text{ (quality at expected level)}$$

$$H_A: \pi \neq \pi_0 \text{ (either better or worse quality than expected, e.g., some event in the production process changed the 'historical' quality level – either for better or for worse)}$$

1. Set confidence level $(1-\alpha) \times 100\%$
2. Choose the sample size n
3. Draw sample and determine the test statistic $P = d/n$
4. Calculate the double sided $(1-\alpha) \times 100\%$ -CI for π : $]\pi_L, \pi_U[$
5. If π_0 belongs to the CI, accept H_0 else reject H_0

One sided test for π

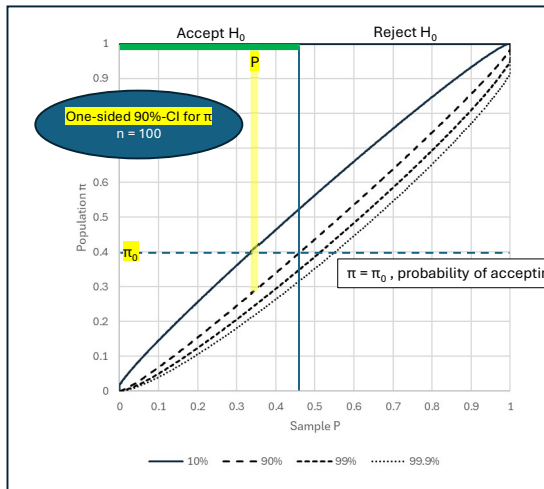
$$H_0: \pi = \pi_0 \text{ (quality at expected level)}$$

$$H_A: \pi < \pi_0 \text{ (better quality than expected, e.g., due to some investment)}$$

1. Set confidence level $(1-\alpha) \times 100\%$
2. Choose the sample size n
3. Draw sample and determine the test statistic $P = d/n$
4. Calculate the one sided $(1-\alpha) \times 100\%$ -CI for π : $[0, \pi_U[$
5. If π_0 belongs to the CI, accept H_0 else reject H_0

Confidence interval

Significance of the test $H_0: \pi \leq \pi_0$ vs. $H_A: \pi > \pi_0$



Significance of the test = 10%

In general:

When testing with a $(1-\alpha) \times 100\%$ -CI, the significance of the test is α .

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We perform the hypothesis test by calculating the $(1-\alpha) \times 100\%$ -CI for π (on the graph we have chosen a confidence level of 90% - so discard the other curves for now).

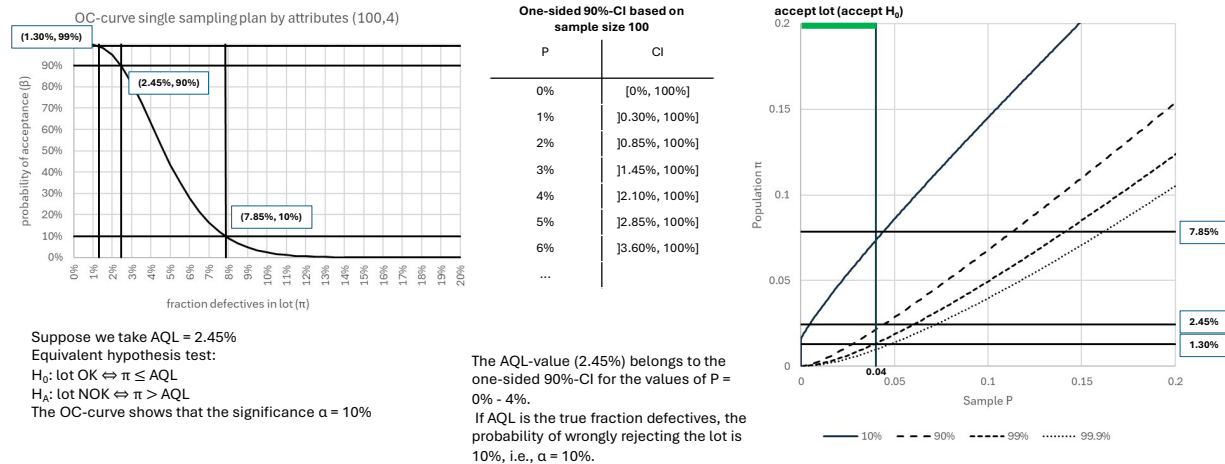
Assume now that H_0 holds, i.e., assume that π (the true fraction defectives) = π_0 . As the true π belongs to the CI in 90% of the samples, we will accept H_0 in 90% of the cases. Hence the significance of the test (the probability of committing the type-I error and wrongly rejecting H_0) is 10%.

We conclude: when we test using a $(1-\alpha) \times 100\%$ -CI, then the significance of the test is α .

The next slide now completes the picture. Testing with a CI is fully equivalent to classical testing, i.e., has the same OC-curve.

Confidence interval

Confirmation of equivalence to classical testing



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So here we show that testing with a CI leads to the same OC-curve as classical testing. We do this by confirming three points of the OC-curve. In the same way any other point of the curve can be confirmed.

Based on binomial probabilities.

The graph on the left shows the OC-curve of a single stage acceptance sampling procedure (100, 4). On the graph are marked 3 points, with respective acceptance probabilities (from low to high): 10%, 90% and 99%. The corresponding fraction defectives in the lot (lot assumed large) are: 7.85%, 2.45% and 1.30%.

A lot with a fraction defectives $\pi = 2.45\%$ has 90% probability to pass the test. If we take 2.45% as the AQL-value, then the procedure can be stated as a hypothesis-test with:

H_0 : the lot is OK for intended use $\Leftrightarrow \pi \leq \text{AQL} = 2.45\%$

H_A : the lot is NOK for intended use $\Leftrightarrow \pi > \text{AQL} = 2.45\%$

The significance α of this test is then 10%. This is the limit case: a lot with precisely this fraction defectives has 10% to be wrongly rejected. A lot with lower fraction defectives, has less chance of being wrongly rejected, e.g., only 1% for a lot with 1.30% defectives.

Lots with a fraction defectives that is higher than 2.45%, have a probability of being wrongly accepted, e.g., for a lot with 7.85% defectives, the probability of being wrongly accepted is 10%.

When drawing a random sample with size 100 from the lot, we can calculate the 90%-CIs for the various outcomes in the sample. For a lot with fraction defectives equal to $\text{AQL} = 2.45\%$, we note that the CI contains the AQL-value for $P = 0\% - 4\%$. These are precisely the values for which we accept the lot (and hence accept H_0).

The figure on the right is a zoom of the figure on the previous page. This figure confirms the three points of the OC-curve mentioned above.

Example t-test for the mean

Case description

A manufacturing process produces parts with a critical dimension that is normally distributed around the value of 10.

After replacement of a broken machine component, management suspects that parts have slightly become smaller, increasing the risk of sending bad parts to the customer.

A careful inspection of 20 randomly selected parts, leads to these data :

10.06 – 9.89 – 9.90 – 9.99 – 9.87 –
9.88 – 9.93 – 9.98 – 10.12 – 9.86 –
9.89 – 9.81 – 9.95 – 9.92 – 10.07 –
9.98 – 10.02 – 9.62 – 9.96 – 9.86

Is management's fear justified?

Test procedure – calculate one sided CI for μ

H_0 : the process is still well centered ($\mu = 10$) (*)

H_A : the process has shifted down ($\mu < 10$)

$\alpha = 2\%$ - confidence level = 98%

rather conservative choice, only adjust process if evidence is strong

test statistic = $\frac{\bar{X} - \mu}{s/\sqrt{n}}$ - t-distributed with 19 df

$\Rightarrow$ one sided 98%-CI for μ

$$=] -\infty, \bar{X} - t_{0.02, 19} \frac{s}{\sqrt{n}} [$$

$$=] -\infty, 9.98 [$$

we observe that $10 \notin \text{CI}$

conclusion

The 98%-CI for the new value of the process center does not contain the previous center value

hence

H_0 cannot be accepted at 2% significance, the process needs adjustment

* In fact $\mu \geq 10$: we are not interested in a possible size increase.

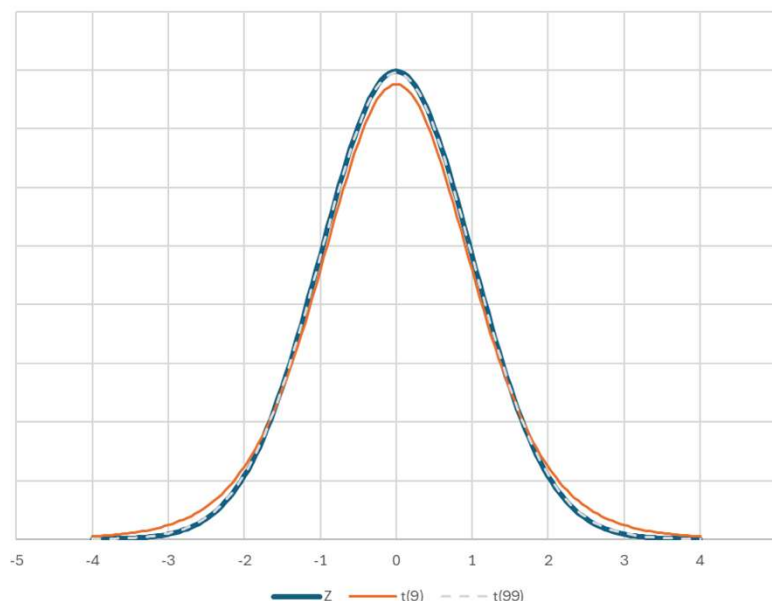
15

See the excel-file 'Black Belt in Six Sigma – Confidence Intervals.xlsx'. We assume in this example that only the process mean is (potentially) affected by the maintenance intervention. The process variability remains at the same level.

We know that when a quantity X is normally distributed, with mean μ and standard deviation σ , then the quantity $Z = \frac{X - \mu}{\sigma}$ is normally distributed with mean 0 and standard deviation 1, notation $Z \sim N(0,1)$.

When we have a sample of values (size n) of X , then the value $t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ follows a so-called t-distribution with $n - 1$ degrees of freedom. The t-distribution is similar to the Z-distribution. It is bell-shaped, symmetrical and has mean 0. Its standard deviation is however somewhat larger than 1. When $n \rightarrow \infty$ th

Z vs. t(df)



Example χ^2 -test for the standard deviation

Case description

A manufacturing process produces parts with a critical dimension that is normally distributed around a nominal value with standard deviation of 0.01.

Due to wear of a machine component, management suspects that the process has become slightly more erratic, increasing the risk of sending bad parts to the customer.

A careful inspection of 20 randomly selected parts, leads to these data :

9.997 – 10.004 – 10.010 – 9.991 – 10.008 –
9.981 – 9.984 – 9.993 – 9.990 – 9.997 –
9.996 – 9.977 – 9.999 – 9.981 – 10.000 –
10.010 – 10.005 – 10.011 – 9.997 – 10.022
Is management's fear justified?

Test procedure

H_0 : process variability has not changed ($\sigma = 0.01$) (*)

H_A : process has become more erratic ($\sigma > 0.01$)

$\alpha = 2\%$ - confidence level = 98%

rather conservative choice, only adjust process if evidence is strong

test statistic = $\frac{(n-1)s^2}{\sigma^2}$ - χ^2 -distributed with 19 df

$\Rightarrow$ one sided 98%-CI for σ

$$=] \sqrt{\frac{19s^2}{\chi_{0.98,19}^2}}, +\infty [$$

$$=] 0.0088, +\infty [$$

we observe that $0.01 \in \text{CI}$

* In fact $\sigma \leq 0.01$: we are not interested in a possible variability decrease.

conclusion

The 98%-CI for the new value of the process standard deviation contains the previous value

hence

H_0 cannot be rejected at 2% significance, the process needs no adjustment

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See the excel-file 'Black Belt in Six Sigma – Confidence Intervals.xlsx'. In this case we assume the process mean to remain stable.

When we have a sample of values (size n) of a normally distributed quantity X with standard deviation σ , then the value $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$ follows a so-called χ^2 -distribution with $n - 1$ degrees of freedom. The χ^2 -distribution is an asymmetric distribution taking only positive values. When $n \rightarrow \infty$ then $\chi^2(n) \rightarrow N(n, \sqrt{2n})$.

χ^2 -density functions

